

Publication Atlas, Caption Contract, and Scholarship Audit

The publication layer is generated from the same typed package state as the stimuli. It contains 15 scientific figures, 10 machine-readable tables, and a separately provenance-recorded editorial cover. The cover is an illustration of the project's subject and methods aesthetic; it is not a stimulus, a participant result, or evidence of a perceptual effect.

Claim-level visualization I

Claim-level visualization I

Claim-level boundary

Each stage exposes a different kind of evidence; later stages do not retroactively validate perception.

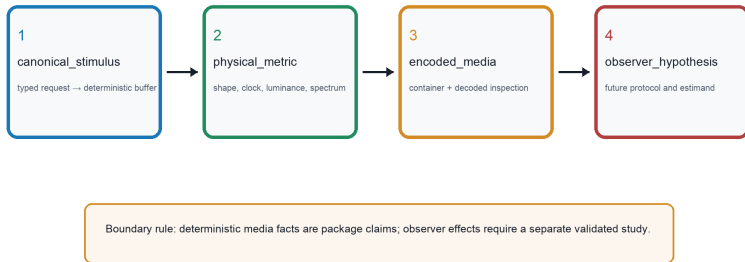


Figure 1: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study.

Claim-level visualization I

What this figure shows: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study. The four-stage boundary distinguishes what DuckRabbit can establish directly—canonical stimulus identity, physical/media metrics, and decoded encoded-media facts—from a future observer hypothesis. Source data `output/data/claim_boundary.json` records the stage definitions and boundary rule; the arrows describe increasing evidential requirements, not an inference that one stage establishes the next. Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram. Objective facts: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar. Claim level: `source_supported`. Source data: `output/data/claim_boundary.json` (SHA-256 digest recorded in the figure

Claim-level visualization I

registry). Evidence lineage: formalism:eq:observer_estimand, formalism:eq:canonical_digest. Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data. Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it.

The claim boundary in fig. 1 is a typed epistemic interface. A canonical digest identifies a deterministic buffer; objective metrics identify properties of that buffer; encoded-media verification identifies facts of the delivered file. A future observer hypothesis is a different object with its own protocol, estimand, and uncertainty interval.

Scholarship map and lineage I

Scholarship map and lineage I

Source-tiered scholarship map

18 catalog entries; 36 source records; exact claims, engineering bases, and limitations are retained in source data.

catalog entry	primary	review	theory	supported claim	status
visual.duck_sabbit	1	1	0	The entry instantiates a controllable antipodal figure structure family. Annotates including: The generation of the geometric performance, not feature reports.	implemented
visual.simultaneous_content	0	1	0	The physical center patches are exhibited while surround luminance differs. Annotates recorded: Clarity reduction and observer adaptation do not manifest.	implemented
visual.apparent_motion	1	1	0	The sequence contains controlled successive spatial events. Annotates recorded: The edited sequences required hardware, not post-processed images.	implemented
audio.stepwand_tavo	1	8	0	The audio contains octave-related patterns with controlled sweep parameters. Annotates recorded: Physical feedback and listener pilot judgments are external.	implemented
audio.missing_fundamental	0	1	0	The normal fundamental component is absent from the canonical spectrum. Annotates recorded: After compensation an observer demonstrates.	implemented
audiovisual.sound_induced_flash	1	1	0	The declared brief/flash event corollis and offsets are physically evoked. Annotates recorded: The stimulus does not contain a visible flash itself.	implemented
audiovisual.ventriloquist	0	2	0	The audio and visual channels carry a declared spatial discrepancy. Annotates recorded: Perceived localization depends on scene, headphones, display, and observer.	implemented
visual.ruler_bar	0	1	1	The two bar lengths are equal in the canonical raster while using geometry vectors. Annotates recorded: The chosen normalized geometry is an engineering variant.	implemented
visual.poggenhoff	0	1	1	The occluder and diagonal combination are generated from explicit geometry. Annotates recorded: Alignment judgments and transfer flows are not measured.	implemented
visual.potato	1	1	0	Target bars are physically equal and ratio converge toward a satisfying region. Annotates recorded: Perceptual discrimination is observed, and observer dependent.	implemented
visual.kanizsa_triangle	1	1	0	The image contains incomplete inducers with no explicitly drawn target edge. Annotates recorded: Illusion contour completion is not directly measured.	implemented
visual.elongation	2	8	0	Control target geometry is held equal while contextual circle geometry differs. Annotates recorded: Perceived size depends on viewing angle and context.	implemented
visual.zollner	2	1	0	The entry instantiates a controlled crossing the orientation tilt/corollis family. Annotates recorded: The ruler exhibits no geometric, non-geometric, or a planar observer perspective.	implemented
audio.tritone_paradox	2	8	0	The canonical pair has an explicit full octave frequency relation. Annotates recorded: Acoustic/physiological reports are not verified before and context.	implemented
audio.achse_illusion	1	8	0	The two channels receive alternating octave-related tones. Annotates recorded: The auditory percept depends on selected presentation and device.	implemented
audio.auditory_continuity	2	8	0	The canonical audio contains a reproduces interruption interval and masker family. Annotates recorded: The generation reports an observed interruption and longer, not a shorter, interval, and no auditory or visual disturbance effect.	implemented
audiovisual.temporal_ventriloquism	2	2	0	Audio and video event timing and declared offset are deterministic, and perceptible. Annotates recorded: The generation experiment pair is a controlled perceptual effect.	implemented
audiovisual.mcgulik	1	8	0	The catalog identifies a speech-dependent audiovisual family without claiming implementation. Annotates recorded: Technical concept report is a perceptual relation context is needed.	input_required

Scholarship map and lineage I

What this figure shows: Rows map catalog entries to source tiers, exact supported claims, engineering bases, limitations, and implementation status. The map links every catalog entry to its primary, review, and theory records, then preserves the exact source-supported claim, engineering departure, limitation, and implementation status. Source data output/data/scholarship_map.json are generated from the checked-in evidence matrix; a source record supports only the statement written in that record and does not certify pixel- or task-identical replication. Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence_matrix.json. Objective facts: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap status. Claim level: source_supported. Source data: output/data/scholarship_map.json

(SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, brugger1999duckrabbit, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist. Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested. Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept.

The source map fig. 2 and tbl. 2 make five distinctions explicit: a primary demonstration, a review, a theoretical account, DuckRabbit's engineering basis, and an evidence limitation. Brugger's duck/rabbit study documents variation across figure variants and observers; this supports a careful ambiguity-family record, not a universal bistability claim (Brugger 1999). Yildiz et al. review competing explanations of Ponzo-like illusions, so the catalog retains both a geometric construction and an unresolved theoretical boundary (Yildiz et al. 2022). Repp's tritone analysis likewise motivates listener- and context-sensitive limitations (Repp 1997).

The audit also treats software and its metadata as part of the scholarly record. FAIR guidance emphasizes machine-actionable provenance and reuse, and software-citation principles emphasize crediting an identifiable version rather than citing an undifferentiated project name (Wilkinson et al. 2016; Smith et al. 2016). Accordingly, the release includes CITATION.cff, CodeMeta, Zenodo metadata, versioned manifests, source-data sidecars, and resolver-linked bibliography entries. These improve discoverability and attribution; they do not increase the evidential level of any perceptual claim.

The public software identity is explicit: DuckRabbit is published and citable at the DuckRabbit public GitHub repository under the authorship of Daniel Ari Friedman, Active Inference Institute. The generated bundle in this repository remains the authoritative release candidate until the external template handoff is completed.

Formal traceability and objective metrics I

Formal traceability and objective metrics I

Formalism traceability

Equation labels resolve to implementation modules, tests, figures, and claim levels.

label	equation	implementation	tests	figures
eq_typed_request claim: canonical_stimulus	$q = \langle \emptyset, \emptyset, s, e \rangle$	registry.py, schema.py	test_generators.py	\$fig architecture, \$fig formalism_traceability
eq_canonical_generation claim: canonical_stimulus	$A = G(B, s)$	registry.py, generators.py	test_generators.py, test_v03_contacts.py	\$fig architecture, \$fig visual_panel
eq_canonical_digest claim: physical_metric	$c(A) = \text{Serialize_LE}(\text{float32}(A, \text{shape}, \text{clock_units});$ $h_c(A) = H(c(A))$	canonical.py, manifest.py	test_v03_contacts.py	\$fig architecture, \$fig encoding_verification
eq_encoding_verification claim: encoded_media	$F = E(A), I = D(F), V(F, M) :: \{\text{pass}, \text{fail}\}$	render.py, inspection.py, manifest.py	test_media_and_render.py, test_v03_contacts.py	\$fig encoding_verification
eq_clock_definition claim: physical_metric	$N = \text{round}(\lfloor s \cdot T \rfloor, t_k = M \cdot t_r, \Delta, AV = t_{\text{audio}} - t_{\text{video}})$	parameters.py, artifacts.py	test_parameters.py, test_media_and_render.py	\$fig temporal_sequence, \$fig audiovisual_timeline
eq_objective_statistics claim: physical_metric	$\mu, Y, \sigma, Y, RMS_X = M(A, \text{units}, \text{tolerance}, \text{version})$	metrics.py, publication.py	test_v04_scholarly.py	\$fig metrics_dashboard, \$fig audio_signals
eq_temporal_spectral_metrics claim: physical_metric	$D_X = \text{mean absolute difference of adjacent frames};$ $\text{centroid}(X) = \sum P_X / \sum P_X$	metrics.py, artifacts.py	test_v04_scholarly.py	\$fig temporal_sequence, \$fig metrics_dashboard
eq_observer_estimand claim: observer_hypothesis	$\psi = E[Y \text{ conditional on condition and protocol}]$	observer_analysis.py, observer.py	test_v04_scholarly.py	\$fig synthetic_psychophysics, \$fig observer_protocol
eq_synthetic_observer claim: synthetic_model_output	$p_M(c A_r, A_c) = \alpha(w \cdot \psi(A_c) - w \cdot \psi(A_r))^\beta$	synthetic_psychophysics.py, metrics.py	test_synthetic_psychophysics.py	\$fig synthetic_psychophysics

Formal traceability and objective metrics I

What this figure shows: Rows connect equation labels to their mathematical definition, implementation paths, tests, figure labels, and claim levels. The traceability registry connects the nine numbered equations to symbols, implementation modules, tests, and registered figures. Source data: `output/data/formalism_traceability.json` is the machine-readable crosswalk used by the manuscript; it demonstrates contract coverage and test linkage without converting formal notation into empirical evidence. Controls: formalism registry version; seed not applicable to the traceability diagram. Objective facts: nine equation records with implementation, test, figure, and claim-level fields; no unit-bearing measurement is plotted. Claim level: `physical_metric`. Source data: `output/data/formalism_traceability.json` (SHA-256 digest recorded in the figure registry). Evidence lineage: `formalism:eq:canonical_digest`, `formalism:eq:clock_definition`,

Formal traceability and objective metrics I

formalism: `eq:objective_statistics`. Limitations: Traceability records documentation and verification scope; it does not establish the truth of an observer-level theory. Boundary: A linked equation and test can show an implemented contract, not a validated perceptual law.

The formalism registry `fig. 3` links equations `eqns. 1, 2, 4, 5, 3, 6, 7, 8, 9` to implementation modules, tests, and figures. The contract is intentionally auditable: equation labels point to code-owned records, while empirical claims remain outside the generator's authority.

Formal traceability and objective metrics I

Objective metrics dashboard

Values are deterministic properties of canonical artifacts; units and computation provenance are explicit.

image

mean_luminance: 0.7770 normalized_luminance
unique_values: 3 levels
effective_dynamic_range: 1.0000 normalized_luminance

audio

rms: 0.1128 normalized_amplitude
peak: 0.3480 normalized_amplitude
spectral_centroid_hz: 1699.5395 Hz
dominant_frequency_hz: 614.0000 Hz

video

frames: 6 frames
frame_rate: 12.0000 frames/second
duration_seconds: 0.5000 seconds
mean_temporal_delta: 0.2422 normalized_pixel_difference

audiovisual

duration_seconds: 0.5000 seconds
sync_offset_ms: 0.0000 milliseconds
spatial_offset: 0.0000 normalized_spatial_offset
audio_rms: 0.0824 normalized_amplitude

Figure 4: Four artifact cards list image, audio, video, and audiovisual metrics with units, computation provenance, and claim level.

Formal traceability and objective metrics I

What this figure shows: Four artifact cards list image, audio, video, and audiovisual metrics with units, computation provenance, and claim level. The dashboard presents representative measurements from image, audio, video, and audiovisual canonical artifacts. Source data output/data/metrics_dashboard.json retains metric name, value, unit, computation version, tolerance, claim level, and canonical digest; luminance is normalized, amplitude is normalized, spectra are in hertz, frame differences are normalized pixel differences, and synchronization is in milliseconds. Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0. Objective facts: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel difference, ms, counts, and durations. Claim level: physical_metric. Source data: output/data/metrics_dashboard.json (SHA-256 digest recorded in the

figure registry). Evidence lineage: formalism:eq:objective_statistics, formalism:eq:temporal_spectral_metrics. Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data.

The metrics dashboard fig. 4 reports units, computation version, and canonical digests. Luminance statistics are normalized raster properties; RMS and peak are normalized-amplitude properties; spectral values are in Hz; frame deltas are normalized pixel differences; and synchronization offsets are in milliseconds. None is a psychophysical score.

Formal traceability and objective metrics I

Typed parameter domains

Intervals are validation domains, not claims about perceptual thresholds or sensitivity.

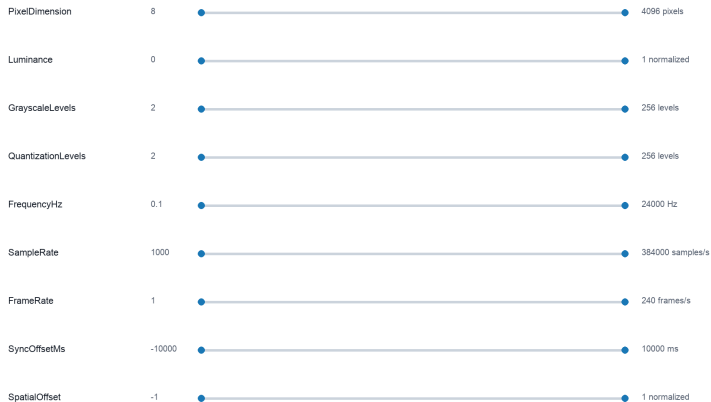


Figure 5: Horizontal interval markers show typed domains for dimensions, luminance, levels, frequency, sampling, frame rate, synchronization, and spatial discrepancy.

Formal traceability and objective metrics I

What this figure shows: Horizontal interval markers show typed domains for dimensions, luminance, levels, frequency, sampling, frame rate, synchronization, and spatial discrepancy. The domain map shows the validated ranges for dimensions, luminance, grayscale and quantization levels, frequency, sampling rate, frame rate, synchronization offset, and spatial discrepancy. Source data output/data/parameter_domains.json records the type name, unit, interval, and engineering role; intervals prevent malformed media and are not proposed as sensitivity thresholds. Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram. Objective facts: dimensionless, pixel, level, Hz, samples/s, frames/s, ms, and normalized spatial units as listed per parameter. Claim level: canonical_stimulus. Source data: output/data/parameter_domains.json (SHA-256 digest recorded in the

Formal traceability and objective metrics I

figure registry). Evidence lineage: formalism:eq:typed_request.
Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a reproducible construction request, not evidence that the requested value is perceptually effective.

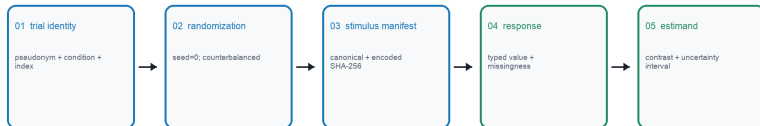
The domain map fig. 5 is a validation visualization. Bounds are chosen to prevent malformed media and unsafe encodings, not to predict a viewer, listener, or participant's sensitivity.

Observer-design boundary I

Observer-design boundary I

Observer protocol: orchestration boundary

The same stimulus manifest can feed a synthetic diagnostic now or a preregistered human study later; the data streams remain separate.



Two explicitly different endpoints



Model templates for future human analysis

logistic_mixed: binomial / logit

lognormal_mixed: lognormal / identity

transition_rate: survival or transition-rate / hazard

linear_mixed: Gaussian / identity

poisson_or_ordinal: Poisson/negative-binomial/ordinal / log or cumulative

No participant records, fitted coefficients, or observer-level result is bundled; the included endpoint is model output only.

Figure 6: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary.

Observer-design boundary I

What this figure shows: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary. The study-ready scaffold moves from pseudonymous trial identity and deterministic randomization to a stimulus manifest and encoded-file hash, typed response or missingness, and a preregistered estimand. Source data output/data/observer_protocol.json records the study design, randomization seed, model templates, and synthetic-only status; no participant record is included. Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data. Objective facts: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable. Claim level: observer_hypothesis. Source data: output/data/observer_protocol.json

Observer-design boundary I

(SHA-256 digest recorded in the figure registry). Evidence lineage: observer:preregistered_estimands. Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol is ready for ethical and preregistered extension, not evidence that the proposed effect exists.

The observer scaffold fig. 6 starts with a pseudonymous trial identity and deterministic randomization, binds the stimulus manifest and encoded hash, records a typed response or missingness state, and ends at a preregistered estimand. The companion diagnostic fig. 9 is a deterministic, hand-specified feature observer with `training_data=none` and `human_data=false`; it is not a pretrained vision model, psychometric function, or participant result. Human psychophysics

Observer-design boundary I

remains a future evidence layer requiring calibrated playback, consent, preregistration, and observed responses.

Generated audit tables I

Table 1: Code-owned caption, source-data, and accessibility audit.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:architecture; canonical stimulus	output/data/architecture.json formalism:eq:typed request, formalism:eq:canonical generation, formalism:eq:encoding verification	Controls: typed request $q=(i, s, e)$ with seed $s=0$; default generator and delivery-independent canonicalization Objective: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar	Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The pipeline verifies reproducible media facts, not what a person experiences. Accessibility: Stage names, mathematical symbols, and arrow direction are printed directly; color is redundant.
fig:catalog matrix; source supported	output/data/catalog matrix.json; gregory1997visual, hirst2020sound, bruns2019ventriloquist	Controls: live taxonomy registry; evidence matrix snapshot; seed not applicable to the catalog diagram Objective: 18 catalog rows; categorical facets and statuses; no observer-level quantity is applicable	Limitations: Taxonomy facets are engineering classifications and remain provisional where the literature supports competing accounts. Boundary: Catalog coverage is bounded to this registered package and does not enumerate all known illusions. Accessibility: Every status and facet is written as text; color only reinforces the printed status.
fig:visual panel; canonical stimulus	output/data/visual panel.json; gregory1997visual, brugger1999duckrabbit, howe2005muller,	Controls: one default typed parameter object per implemented visual entry; seed $s=0$; dimensions, luminance bounds,	Limitations: The gallery is complete for implemented visual entries in this package, not a complete survey of visual illusions;

Generated audit tables II

fig:visual sweep; physical
metric

output/data/visual
sweep.json;
brugger1999duckrabbit,
gregory1997visual

Controls: duck weight
{0,.25,.5,.75,1};
grayscale=quantization
{2,4,8,16,32}; seed s=0
Objective: 25 canonical
rasters; unique levels,
normalized mean luminance,
and effective dynamic range
per cell

Limitations: The grid is a
sensitivity of the
construction parameters,
not a psychophysical
sensitivity curve or a
validated observer effect.
Boundary: The chosen grid
is an engineering sampling
of the parameter domain
and does not imply an
optimal or perceptually
uniform spacing.
Accessibility: Row and
column labels state the
parameter values; numerical
metrics and units are
available in the sidecar.
Limitations: Frame
succession and frame deltas
are physical file properties;
playback timing, display
persistence, and motion
reports require a controlled
observer task. Boundary:
The figure documents
temporal structure rather
than the presence, direction,
or strength of perceived
motion. Accessibility:
Frame index, seconds,
frames per second, and
normalized-difference labels
remain interpretable
without color.

fig:temporal sequence;
physical metric

output/data/temporal
sequence.json;
wertheimer1912motion,
sekuler1996wertheimer

Controls: visual.apparent
motion default typed
parameters; frame rate and
frame count from the
canonical sequence; seed
s=0 Objective: timestamps
in seconds, frame rate in
frames/s, frame count,
rational timebase, and
adjacent-frame normalized
pixel difference

Generated audit tables III

fig:audio signals; physical metric	output/data/audio signals.json; shepard1984scale, zatorre2005missing, deutsch1986tritone, repp1997tritone, deutsch1974octave, warren1970continuity, riecke2011continuity	Controls: default typed audio parameters; per-generator channel layout and sample rate; seed s=0; one-sided FFT display with 48 sampled bins Objective: RMS and peak in normalized amplitude; spectral centroid and bandwidth in Hz; sample rate in samples/s; channel layout and canonical SHA-256 digest	Limitations: Playback level, transducer response, dichotic separation, masker design, and listener judgments are external to the canonical buffer. Boundary: The figure compares generated signal properties and literature-linked families; it does not establish pitch, continuity, or stream organization for a listener. Accessibility: Waveform, spectral, amplitude, and frequency labels are printed; bar color is not the sole encoding.
fig:audiovisual timeline; physical metric	output/data/audiovisual timeline.json; shams2000sifi, hirst2020sound, vroomen2004temporal, hartcherobrien2011temporal	Controls: audiovisual.sound induced flash default shared timeline; declared sync offset ΔAV and spatial offset; seed s=0 Objective: video luminance and audio envelope on normalized shared time; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidebar	Limitations: Display refresh, audio hardware, event latency, and observer temporal binding are not measured by this figure. Boundary: A shared-clock construction is not evidence that observers bind the events at the declared or any other perceptual time. Accessibility: The two channels, time axis, signed offset convention, and normalized spatial discrepancy are labeled in text.

Generated audit tables IV

fig:encoding verification;
encoded media

output/data/encoding
verification.json;
engineering:typed media
adapters

Controls: typed encoding
profile selected per artifact;
overwrite and backend
capability checks; seed
inherited from the stimulus
request Objective: format
availability and verification
scope are environment
observations; NPZ is exact
while other formats are
decoded and
tolerance-checked

Limitations: Backend
availability, codec versions,
container metadata, and
lossy tolerances are
environment-dependent;
playback software can
expose additional behavior.
Boundary: A successful
encode or decode does not
validate an observer effect
or guarantee cross-device
perceptual equivalence.
Accessibility: Format,
backend, preserved fact,
and verification columns are
textual and remain usable
without color.
Limitations: The model is
hand-specified, has no
training data, has not been
calibrated against observers,
and does not estimate a
human threshold or effect
size. Its probabilities are
model outputs, not
participant responses or a
psychometric function.
Boundary: This diagnostic
tests deterministic
orchestration and
metamorphic input-output
behavior only; empirical
psychophysics remains
future work. Accessibility:
Model scale, reference line,
model identity, and human d

fig:synthetic psychophysics;
synthetic model output

output/data/synthetic
psychophysics.json;
observer:preregistered
estimands

Controls: duck weight
comparison against fixed
reference; serialized feature
weights and temperature;
seed s=0 Objective: model
probability and score delta
on the displayed
model-output scale;
canonical stimulus digests;
human data=false; training
data=none

Generated audit tables V

fig:claim boundary; source supported	output/data/claim boundary.json; formalism:eq:observer estimand, formalism:eq:canonical digest	Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram Objective: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar	Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data. Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it. Accessibility: Each stage is named, numbered, and described in text; the boundary rule is readable without color.
fig:scholarship map; source supported	output/data/scholarship map.json; gregory1997visual, brugger1999duckrabbitt, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist	Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence matrix.json Objective: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap status	Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested. Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept. Accessibility: Source roles, claims, limitations, and statuses are printed or available in the sidecar; no category is encoded by color alone.
fig:formalism traceability; physical metric	output/data/formalism traceability.json; formalism:eq:canonical	Controls: formalism registry version; seed not applicable to the traceability diagram	Limitations: Traceability records documentation and verification scope; it does

Generated audit tables VI

fig:metrics dashboard; physical metric	output/data/metrics dashboard.json; formalism:eq:objective statistics, formalism:eq:temporal spectral metrics	Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0 Objective: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel difference, ms, counts, and durations	Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data. Accessibility: Every numerical value is paired with a printed unit or an explicit dimensionless definition.
fig:parameter domains; canonical stimulus	output/data/parameter domains.json; formalism:eq:typed request	Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram Objective: dimensionless, pixel, level, Hz, samples/s, frames/s, ms, and normalized spatial units as listed per parameter	Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a reproducible construction request, not evidence that the requested value is perceptually effective. Accessibility: Type names, endpoints, units, and roles are printed; interval color is redundant.

Generated audit tables VII

fig:observer protocol; observer hypothesis	output/data/observer protocol.json; observer:preregistered estimands	Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data Objective: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable	Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol is ready for ethical and preregistered extension, not evidence that the proposed effect exists. Accessibility: Every protocol step and the no-participant-data boundary is stated in text; arrows and labels remain legible without color.
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Table 2: Source-tiered evidence, DOI/URL verification, exact supported claims, engineering departures, limitations, and explicit planned/input-required gaps.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
gregory 1997 visual	review; gregory 1997 visual	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	A non-exhaustive classification of visual illusion families. A review of sound-induced flash paradigms and multisensory temporal inference. A statistical image-source account of Müller-Lyer geometry. A mechanistic account of Poggendorff orientation estimation. A primary investigation of identical targets embedded in an angular context. Canonical illusory-contour inducer arrangements. Context-dependent size judgments in Ebbinghaus-family stimuli. Foundational apparent-motion experiments using successive spatial events.
hirst 2020 sound	review; hirst 2020 sound	DOI: recorded; URL host/path: www.sciencedirect.com	snapshot validated; offline snapshot	
howe 2005 muller	theory; howe 2005 muller	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
morgan 1999 pogendorff	theory; morgan 1999 pogendorff	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
fisher 1967 ponzo	primary; fisher 1967 ponzo	DOI: recorded; URL host/path: www.nature.com	snapshot validated; offline snapshot	
kanizsa 1976 contours	primary; kanizsa 1976 contours	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
mruczek 2015 ebbinghaus	primary; mruczek 2015 ebbinghaus	DOI: recorded; URL host/path: pmc.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
wertheimer 1912 motion	primary; wertheimer 1912 motion	DOI: not recorded; URL host/path: bibbase.org	snapshot validated; offline snapshot	

Generated audit tables IX

sekuler 1996 wertheimer	review; sekuler 1996 wertheimer	DOI: recorded; URL host/path: journals.sagepub.com	snapshot validated; offline snapshot	A review connecting Wertheimer's successive-event findings to later apparent-motion research and clarifying their historical scope. Auditory scale and pitch-class organization in Shepard-like tones. The missing-fundamental problem as a pitch-inference phenomenon. The tritone paradox and dependence on pitch-class context and listener. Dichotic alternating high/low tone construction underlying the octave illusion. The one-flash/two-beep sound-induced flash contrast. Spatial audiovisual capture and multisensory integration constraints.
shepard 1984 scale	primary; shepard 1984 scale	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
zatorre 2005 missing	review; zatorre 2005 missing	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
deutsch 1986 tritone	primary; deutsch 1986 tritone	DOI: recorded; URL host/path: online.ucpress.edu	snapshot validated; offline snapshot	
deutsch 1974 octave	primary; deutsch 1974 octave	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
shams 2000 sifi	primary; shams 2000 sifi	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
bruns 2019 ventriloquist	review; bruns 2019 ventriloquist	DOI: recorded; URL host/path: pmc.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	

Generated audit tables X

vroomen 2004 temporal	primary; vroomen 2004 temporal	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Sound timing can alter the apparent temporal structure of a visual event. Speech-dependent audiovisual categorical integration, requiring validated fixtures. Auditory continuity/restoration as a future calibrated masker family. Variation in duck/rabbit ambiguity across figure variants and observers. Competing depth-based and non-depth-based explanations of Ponzo-like illusions. Listener- and context-dependent effects of spectral envelope and pitch-class structure in tritone judgments. Audiovisual timing can shift temporal judgments in a purely temporal context without spatial grounding. Crossing oblique and long-line geometry used to measure orientation
mcgurk 1976 speech	primary; mcgurk 1976 speech	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
warren 1970 continuity	primary; warren 1970 continuity	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
brugger 1999 duckrabbit	primary; brugger 1999 duckrabbit	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
yildiz 2022 ponzo	review; yildiz 2022 ponzo	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
repp 1997 tritone	primary; repp 1997 tritone	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
hartcherobrien 2011 temporal	primary; hartcherobrien 2011 temporal	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
earle 1995 zollner	primary; earle 1995 zollner	DOI: recorded; URL host/path: journals.sagepub.com	snapshot validated; offline snapshot	

Generated audit tables XI

zoellner 1860 pseudoscopy	primary; zoellner 1860 pseudoscopy	DOI: recorded; URL host/path: onlinelibrary.wiley.com	snapshot validated; offline snapshot	The nineteenth-century primary description of the crossing-line orientation family later known as the Zöllner illusion. Interrupted sounds with masking context and the separation of sensory and decisional contributions. Contemporary perceptual-grouping, contour-completion, and figure-ground accounts relevant to Kanizsa-type constructions. A classic contextual-size experiment varying surround geometry and comparison conditions in Ebbinghaus-family displays. Causal-inference and temporal-prediction accounts of audiovisual integration and segregation. The English translation and commentary preserves Ibn al-Haytham's eleventh-century
riecke 2011 continuity	primary; rieke 2011 continuity	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
wagemans 2012 gestalt	review; wagemans 2012 gestalt	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
weintraub 1979 ebbinghaus	primary; weintraub 1979 ebbinghaus	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
noppeney 2018 causal	review; noppeney 2018 causal	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
alhazen 1989 optics	primary; alhazen 1989 optics	DOI: not recorded; URL host/path: www.cca.qc.ca	snapshot validated; offline snapshot	

Generated audit tables XII

mozi 2023 canons	review; mozi 2023 canons	DOI: recorded; URL host/path: link.springer.com	snapshot validated; offline snapshot	A modern translation and commentary documents Mohist Canon sections on knowledge, optics, and mechanics, providing a Chinese Warring States-period context for early technical reasoning about vision.
ptolemy 1996 optics	primary; ptolemy 1996 optics	DOI: not recorded; URL host/path: sites.dlib.nyu.edu	snapshot validated; offline snapshot	The scholarly translation makes Ptolemy's second-century Optics available as an early work on visual perception and mathematical visual theory.
berkeley 1709 vision	primary; berkeley 1709 vision	DOI: not recorded; URL host/path: www.maths.tcd.ie	snapshot validated; offline snapshot	Berkeley's 1709 essay treats visual distance and space as mediated by learned associations rather than as a simple direct readout of visual geometry.
kircher 1646 light	primary; kircher 1646 light	DOI: not recorded; URL host/path: collections.st-andrews.ac.uk	snapshot validated; offline snapshot	The digitized record anchors Kircher's 1646 <i>Ars magna lucis et umbrae</i> as an early-modern work on light, shadow, and optical display.

Generated audit tables XIII

molyneux 2020 problem	review; molyneux 2020 problem	DOI: not recorded; URL host/path: plato.stanford.edu	snapshot validated; offline snapshot	The historical account documents Molyneux's 1688 question separating tactile learning from visual recognition and its long cross-sensory afterlife. The digitized Philosophical Transactions record preserves Cheselden's 1728 case report as a historical observation relevant to visual access and the separation of observer evidence from stimulus description. A history of Chinese science and technology surveys ancient Chinese physics and records optical phenomena as part of a non-European history of visual science. The entry instantiates a controllable ambiguous-figure stimulus family. Engineering basis: Parameterized ambiguous silhouette; not a pixel-identical reproduction of a
cheselden 1728 sight	primary; cheselden 1728 sight	DOI: not recorded; URL host/path: archive.org	snapshot validated; offline snapshot	
dai 2015 chinesescience	review; dai 2015 chinesescience	DOI: recorded; URL host/path: link.springer.com	snapshot validated; offline snapshot	
entry / visual.duck rabbit	engineering / limitation / gap; brugger 1999 duckrabbit; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	

Generated audit tables XIV

entry / visual.simultaneous contrast	engineering / limitation / gap; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The physical center patches are matched while surround luminance differs. Engineering basis: Equal central patches on unequal luminance surrounds. Limitation: Display calibration and observer adaptation are not modeled. Missing contract: none
entry / visual.apparent motion	engineering / limitation / gap; wertheimer 1912 motion; sekuler 1996 wertheimer	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The sequence contains controlled successive spatial events. Engineering basis: Alternating rectangular masks at a fixed rational frame rate. Limitation: The artifact establishes temporal succession, not perceived motion. Missing contract: none
entry / audio.shepard tone	engineering / limitation / gap; shepard 1984 scale	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The audio contains octave-related partials with controlled sweep parameters. Engineering basis: Additive octave-spaced partials with a deterministic envelope. Limitation: Playback transducers

Generated audit tables XV

entry / audio.missing fundamental	engineering / limitation / gap; zatorre 2005 missing	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The nominal fundamental component is absent from the canonical spectrum. Engineering basis: Harmonic components are synthesized while the nominal fundamental is omitted. Limitation: Pitch completion is an observer-level inference. Missing contract: none
entry / audiovisual.sound induced flash	engineering / limitation / gap; shams 2000 sifi; hirst 2020 sound	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The declared beep/flash event counts and offsets are physically encoded. Engineering basis: One flash paired with one or two timed beeps on a shared clock. Limitation: The stimulus does not establish a reported flash count. Missing contract: none
entry / audiovi- sual.ventriloquist	engineering / limitation / gap; bruns 2019 ventriloquist; nopeney 2018 causal	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The audio and visual channels carry a declared spatial discrepancy. Engineering basis: Stereo panning and visual displacement encode a spatial

Generated audit tables XVI

entry / visual.muller lyer	engineering / limitation / gap; gregory 1997 visual; howe 2005 muller	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The two bar lengths are equal in the canonical raster while wing geometry varies. Engineering basis: Equal bars with controlled arrow-wing geometry. Limitation: The chosen normalized geometry is one engineering variant. Missing contract: none
entry / visual.poggendorff	engineering / limitation / gap; gregory 1997 visual; morgan 1999 poggendorff	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The occluder and diagonal continuation are generated from explicit geometry. Engineering basis: A diagonal continuation is separated by a rectangular occluder. Limitation: Alignment judgments and orientation filters are not measured. Missing contract: none
entry / visual.ponzo	engineering / limitation / gap; fisher 1967 ponzo; yildiz 2022 ponzo	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Target bars are physically equal and rails converge toward a vanishing region. Engineering basis: Equal target bars are placed inside converging rails. Limitation: S

Generated audit tables XVII

entry / visual.kanizsa triangle	engineering / limitation / gap; kanizsa 1976 contours; wagemans 2012 gestalt	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The image contains incomplete inducers with no explicitly drawn triangle edge. Engineering basis: Three incomplete circular inducers are rasterized around a triangular gap. Limitation: Illusory contour completion is not directly measured. Missing contract: none
entry / visual.ebbinghaus	engineering / limitation / gap; mruczek 2015 ebbinghaus; weintraub 1979 ebbinghaus	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Central target geometry is held equal while contextual circle geometry differs. Engineering basis: Equal central targets are surrounded by different context-circle sizes. Limitation: Perceived size depends on viewing scale and context. Missing contract: none
entry / visual.zollner	engineering / limitation / gap; zoellner 1860 pseudoscopy; earle 1995 zollner; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The entry instantiates a controlled crossing-line orientation stimulus family. Engineering basis: Vertical long lines crossed by short oblique inducers with

Generated audit tables XVIII

entry / audio.tritone paradox	engineering / limitation / gap; deutsch 1986 tritone; repp 1997 tritone	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The canonical pair has an explicit half-octave frequency relation. Engineering basis: Two octave-complex tones are separated by a half-octave interval. Limitation: Ascending/descending reports vary across listeners and contexts. Missing contract: none
entry / audio.octave illusion	engineering / limitation / gap; deutsch 1974 octave	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The two channels receive alternating octave-related tones. Engineering basis: Alternating high and low tones are assigned to opposite stereo channels. Limitation: The auditory percept depends on dichotic presentation and listener. Missing contract: none
entry / audio.auditory continuity	engineering / limitation / gap; warren 1970 continuity; riecke 2011 continuity	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The canonical audio contains a reproducible interruption interval and masker family. Engineering basis: An interrupted sinusoidal carrier is replaced during a declared gap by a deterministic

Generated audit tables XIX

entry / audiovisual.temporal ventriloquism	engineering / limitation / gap; vroomen 2004 temporal; hartcherobrien 2011 temporal; hirst 2020 sound; noppeney 2018 causal	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Audio and video event timing and declared offset are deterministic and inspectable. Engineering basis: A visual flash and audio click share a clock with explicit offset. Limitation: The generated single-event pair is a simplified engineering stimulus. Missing contract: none
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Generated audit tables XX

entry / audiovisual.mcgurk	engineering / limitation / gap; mcgurk 1976 speech	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The catalog identifies a speech-dependent audiovisual family without claiming implementation. Engineering basis: Requires validated speech audio/video fixtures and licensing records. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Missing contract: A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required.
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Generated audit tables XXI Table 3: Formal equation-to-code traceability.

Label	Definition and symbols	Implementation and tests	Figures and claim level
eq:typed request	$q = (i, , s, e)$ Symbols: $i, , s, e$	Code: registry.py, schema.py; Tests: test generators.py	Figures: fig:architecture, fig:formalism traceability; Level: canonical stimulus
eq:canonical generation	$A = G(; s)$ Symbols: $A, G, , s$	Code: registry.py, generators.py; Tests: test generators.py, test v03 contracts.py	Figures: fig:architecture, fig:visual panel; Level: canonical stimulus
eq:canonical digest	$c(A) = \text{Serialize_LE, float32}(A, \text{shape, clock, units})$; $h_c(A) = H(c(A))$ Symbols: $c(A), h_c, H$	Code: canonical.py, manifest.py; Tests: test v03 contracts.py	Figures: fig:architecture, fig:encoding verification; Level: physical metric
eq:encoding verification	$F = E(A), I = D(F), V(F, M) \{ \text{pass, fail} \}$ Symbols: F, E, D, I, V, M	Code: render.py, inspection.py, manifest.py; Tests: test media and render.py, test v03 contracts.py	Figures: fig:encoding verification; Level: encoded media
eq:clock definition	$N = \text{round}(f_s T), t_k = k/f_r, \Delta_AV = t_audio - t_video$ Symbols: $N, f_s, T, t_k, f_r, \Delta_AV$	Code: parameters.py, artifacts.py; Tests: test parameters.py, test media and render.py	Figures: fig:temporal sequence, fig:audiovisual timeline; Level: physical metric
eq:objective statistics	$_Y, _Y, \text{RMS_X} = M(A; \text{units, tolerance, version})$ Symbols: $_Y, _Y, \text{RMS_X}, M$	Code: metrics.py, publication.py; Tests: test v04 scholarly.py	Figures: fig:metrics dashboard, fig:audio signals; Level: physical metric
eq:temporal spectral metrics	$D_k = \text{mean absolute difference of adjacent frames}$; $\text{centroid}(X) = \Sigma fP_X(f)/\Sigma P_X(f)$ Symbols: D_k, P_X, f	Code: metrics.py, artifacts.py; Tests: test v04 scholarly.py	Figures: fig:temporal sequence, fig:metrics dashboard; Level: physical metric
eq:observer estimand	$= E[Y \text{ conditional on condition and protocol}]$ Symbols: $, Y, \text{condition, protocol}$	Code: observer analysis.py, observer.py; Tests: test v04 scholarly.py	Figures: fig:synthetic psychophysics, fig:observer protocol; Level: observer hypothesis

eq:synthetic observer	$p_M(c \text{ given } A_r, A_c) = \frac{(w \cdot (A_c) - w \cdot (A_r))}{p_M, c, A_r, A_c, w, \dots}$	Code: synthetic psychophysics.py, metrics.py; Tests: test synthetic psychophysics.py	Figures: fig:synthetic psychophysics; Level: synthetic model output
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Table 4: Claim ledger separating code-derived facts, scholarship, engineering departures, limitations, scope, and observer hypotheses.

Claim ID	Claim and level	Basis	Lineage and limitation
catalog:count	The catalog contains 18 entries. Level: canonical_stimulus	derived_from_code	Lineage: taxonomy entries() Limitation: Catalog membership is not a completeness claim about all known illusions. Manuscript: manuscript/09 appendix catalog.md
evidence:source count	The checked-in evidence matrix contains 36 source records. Level: source_supported	checked_in_scholarship	Lineage: data/evidence matrix.json::sources Limitation: The offline snapshot does not imply that every URL is currently reachable or that a source supports more than its exact record. Manuscript: manuscript/06 scope and related work.md
scope:no participant data	No participant data are bundled with the package. Level: observer_hypothesis	derived_from_code	Lineage: experiments/observer protocol.md; output/data/synthetic psychophysics.json Limitation: Future observer studies require preregistration, consent, calibrated presentation, and separate data governance. Manuscript: manuscript/07 publication audit.md
synthetic:diagnostic boundary	The synthetic observer is a deterministic model-output diagnostic, not human psychophysics. Level: synthetic_model_output	synthetic_model_output	Lineage: src/duckrabbit/synthetic psychophysics.py; output/data/synthetic psychophysics.json

Generated audit tables XXIV

cover:editorial boundary	The cover is a publication illustration, not an experimental stimulus or observer result. Level: publication_illustration	publication_illustration	Lineage: output/reports/cover visualization.json Limitation: The editorial asset is not part of the deterministic scientific stimulus registry. Manuscript: manuscript/07 publication audit.md Lineage: sources=brugger1999duckrabbit; gregory1997visual; data/evidence matrix.json; engineering=Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate.; limitation=The generator verifies geometry and luminance, not bistable reports.; missing contract=none Limitation: The generator verifies geometry and luminance, not bistable reports. Manuscript: manuscript/06 scope and related work.md Lineage: sources=gregory1997visual; data/evidence matrix.json; engineering=Equal central patches on unequal luminance surrounds.; limitation=Display calibration and observer adaptation are not modeled; limitation
evidence:visual.duck rabbit	The entry instantiates a controllable ambiguous-figure stimulus family. Level: source_supported	checked_in_scholarship	
evidence:visual.simultaneous contrast	The physical center patches are matched while surround luminance differs. Level: source_supported	checked_in_scholarship	

Generated audit tables XXV

evidence:visual.apparent motion	The sequence contains controlled successive spatial events. Level: source_supported	checked_in_scholarship	Lineage: sources=wertheimer1912motion; sekuler1996wertheimer; data/evidence matrix.json; engineering=Alternating rectangular masks at a fixed rational frame rate.; limitation=The artifact establishes temporal succession, not perceived motion.; missing contract=none Limitation: The artifact establishes temporal succession, not perceived motion. Manuscript: manuscript/06 scope and related work.md Lineage: sources=shepard1984scale; data/evidence matrix.json; engineering=Additive octave-spaced partials with a deterministic envelope.; limitation=Playback transducers and listener pitch judgments are external.; missing contract=none Limitation: Playback transducers and listener pitch judgments are external. Manuscript: manuscript/06 scope and related work.md Lineage: sources=zatorre2005missing; data/evidence matrix.json; engineering=Harmonic
evidence:audio.shepard tone	The audio contains octave-related partials with controlled sweep parameters. Level: source_supported	checked_in_scholarship	
evidence:audio.missing fundamental	The nominal fundamental component is absent from the canonical spectrum. Level: source_supported	checked_in_scholarship	

Generated audit tables XXVI

evidence:audiovisual.sound induced flash	The declared beep/flash event counts and offsets are physically encoded. Level: source_supported	checked_in_scholarship
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Lineage:
sources=shams2000sifi;
hirst2020sound;
data/evidence matrix.json;
engineering=One flash
paired with one or two
timed beeps on a shared
clock.; limitation=The
stimulus does not establish
a reported flash count.;
missing contract=none
Limitation: The stimulus
does not establish a
reported flash count.

Manuscript: manuscript/06
scope and related work.md

evidence:audiovisual.ventriloquist The audio and visual channels carry a declared spatial discrepancy. Level: source_supported	checked_in_scholarship
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Lineage:
sources=bruns2019ventriloquist;
noppene2018causal;
data/evidence matrix.json;
engineering=Stereo panning
and visual displacement
encode a spatial
discrepancy.;
limitation=Perceived
localization depends on
room, headphones, display,
and observer.; missing
contract=none Limitation:
Perceived localization
depends on room,
headphones, display, and
observer. Manuscript:
manuscript/06 scope and
related work.md

Generated audit tables XXVII

evidence:visual.muller lyer	The two bar lengths are equal in the canonical raster while wing geometry varies. Level: source_supported	checked_in_scholarship	Lineage: sources=gregory1997visual; howe2005muller; data/evidence matrix.json; engineering=Equal bars with controlled arrow-wing geometry.; limitation=The chosen normalized geometry is one engineering variant.; missing contract=none Limitation: The chosen normalized geometry is one engineering variant. Manuscript: manuscript/06 scope and related work.md
evidence:visual.poggendorff	The occluder and diagonal continuation are generated from explicit geometry. Level: source_supported	checked_in_scholarship	Lineage: sources=gregory1997visual; morgan1999poggendorff; data/evidence matrix.json; engineering=A diagonal continuation is separated by a rectangular occluder.; limitation=Alignment judgments and orientation filters are not measured.; missing contract=none Limitation: Alignment judgments and orientation filters are not measured. Manuscript: manuscript/06 scope and related work.md
evidence:visual.ponzo	Target bars are physically equal and rails converge toward a vanishing region. Level: source_supported	checked_in_scholarship	Lineage: sources=fisher1967ponzo; yildiz2022ponzo; data/evidence matrix.json; engineering=Equal target bars converging toward a vanishing region.

Generated audit tables XXVIII

evidence:visual.kanizsa triangle	The image contains incomplete inducers with no explicitly drawn triangle edge. Level: source_supported	checked_in_scholarship	Lineage: sources=kanizsa1976contours; wagemans2012gestalt; data/evidence matrix.json; engineering=Three incomplete circular inducers are rasterized around a triangular gap.; limitation=Illusory contour completion is not directly measured.; missing contract=none Limitation: Illusory contour completion is not directly measured. Manuscript: manuscript/06 scope and related work.md Lineage: sources=mruczek2015ebbinghaus; weintraub1979ebbinghaus; data/evidence matrix.json; engineering=Equal central targets are surrounded by different context-circle sizes.; limitation=Perceived size depends on viewing scale and context.; missing contract=none Limitation: Perceived size depends on viewing scale and context. Manuscript: manuscript/06 scope and related work.md Lineage: sources=zoellner1860pseudoscopy; earle1995zollner; gregory1997visual; data/evidence matrix.json;
evidence:visual.ebbinghaus	Central target geometry is held equal while contextual circle geometry differs. Level: source_supported	checked_in_scholarship	Lineage: sources=mruczek2015ebbinghaus; weintraub1979ebbinghaus; data/evidence matrix.json; engineering=Equal central targets are surrounded by different context-circle sizes.; limitation=Perceived size depends on viewing scale and context.; missing contract=none Limitation: Perceived size depends on viewing scale and context. Manuscript: manuscript/06 scope and related work.md Lineage: sources=zoellner1860pseudoscopy; earle1995zollner; gregory1997visual; data/evidence matrix.json;
evidence:visual.zollner	The entry instantiates a controlled crossing-line orientation stimulus family. Level: source_supported	checked_in_scholarship	Lineage: sources=zoellner1860pseudoscopy; earle1995zollner; gregory1997visual; data/evidence matrix.json;

Generated audit tables XXIX

evidence:audio.tritone paradox	The canonical pair has an explicit half-octave frequency relation. Level: source_supported	checked_in_scholarship	Lineage: sources=deutsch1986tritone; repp1997tritone; data/evidence matrix.json; engineering=Two octave-complex tones are separated by a half-octave interval.; limitation=Ascending/descending reports vary across listeners and contexts.; missing contract=none Limitation: Ascending/descending reports vary across listeners and contexts. Manuscript: manuscript/06 scope and related work.md
evidence:audio.octave illusion	The two channels receive alternating octave-related tones. Level: source_supported	checked_in_scholarship	Lineage: sources=deutsch1974octave; data/evidence matrix.json; engineering=Alternating high and low tones are assigned to opposite stereo channels.; limitation=The auditory percept depends on dichotic presentation and listener.; missing contract=none Limitation: The auditory percept depends on dichotic presentation and listener. Manuscript: manuscript/06 scope and related work.md
evidence:audio.auditory continuity	The canonical audio contains a reproducible interruption interval and another form of such	checked_in_scholarship	Lineage: sources=warren1970continuity; riecke2011continuity; data/evidence matrix.json;

Generated audit tables XXX

evidence:audiovisual.temporal
ventriloquism

Audio and video event timing and declared offset are deterministic and inspectable. Level: source_supported

checked_in_scholarship

Lineage:
sources=vroomen2004temporal;
hartchero-
brien2011temporal;
hirst2020sound;
noppenev2018causal;
data/evidence matrix.json;
engineering=A visual flash and audio click share a clock with explicit offset.;
limitation=The generated single-event pair is a simplified engineering stimulus.; missing contract=none Limitation: The generated single-event pair is a simplified engineering stimulus.
Manuscript: manuscript/06 scope and related work.md
Lineage:
sources=mcgurk1976speech;
data/evidence matrix.json;
engineering=Requires validated speech audio/video fixtures and licensing records.;
limitation=No fixture, consent record, or perceptual validation contract is bundled.;
missing contract=A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual

evidence:audiovisual.mcgurk

The catalog identifies a speech-dependent audiovisual family without claiming implementation. Level: source_supported

checked_in_scholarship

Generated audit tables XXXI

publication:figure count	The publication atlas contains 15 registered scientific figures. Level: physical_metric	derived_from_code	Lineage: publication caption specs() Limitation: The count is a package-state fact, not evidence of empirical validity. Manuscript: manuscript/07 publication audit.md
publication:table count	The publication appendix contains the generated table set defined by the publication table payload registry. Level: physical_metric	derived_from_code	Lineage: publication table payloads() Limitation: Generated tables summarize package state and do not replace source or observer evidence. Manuscript: manuscript/07 publication audit.md

Generated audit tables I

The source-tiered evidence table tbl. 2, caption audit tbl. 1, formalism registry tbl. 3, and claim ledger tbl. 4 are generated sidecars. They provide a publication appendix without duplicating mutable counts or silently converting literature statements into package results.

The publication artifacts have two distinct provenance boundaries. Scientific figures are deterministic PNG renders whose source-data JSON and SHA-256 digests are checked against the code-owned registry. The cover is an editorial charcoal illustration: its selected candidate, prompt fingerprint, source asset digest, dimensions, variants, and provenance boundary are recorded in `output/reports/cover_visualization.json`. Neither boundary is an observer result, and neither substitutes for display calibration or behavioral data.

The publication audit additionally resolves every formalism edge to a real package module, test file, manuscript equation label, and registered figure. Figure source-data sidecars use a versioned schema with figure identity, generator, seed, and nested data payload; the registry verifies this identity and hash independently of the renderer. This makes a visually plausible but stale or relabeled figure fail the same provenance gate as a tampered stimulus.